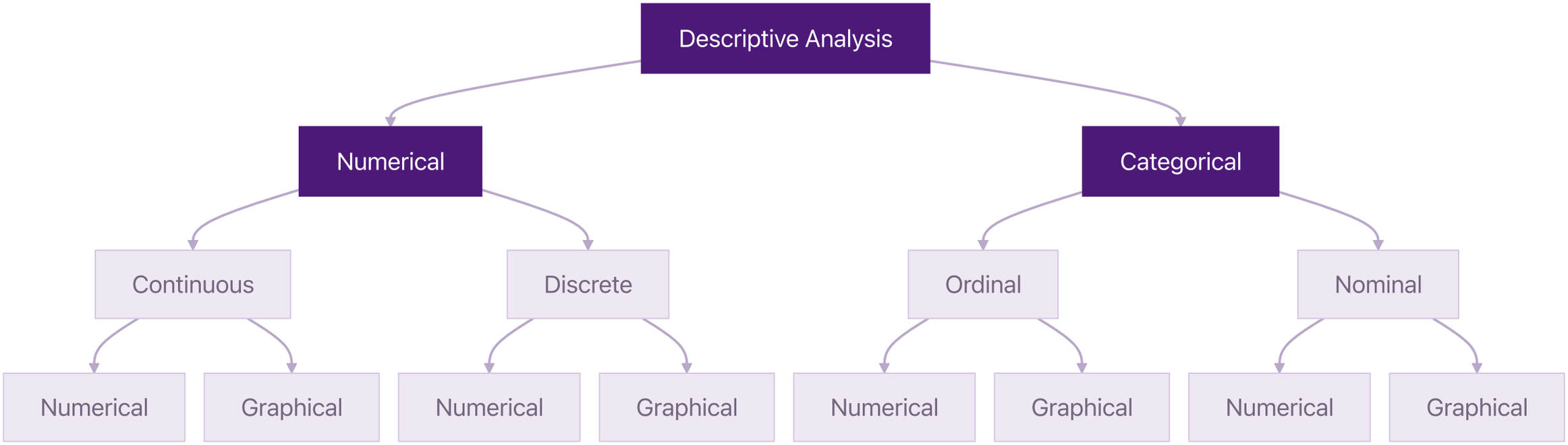
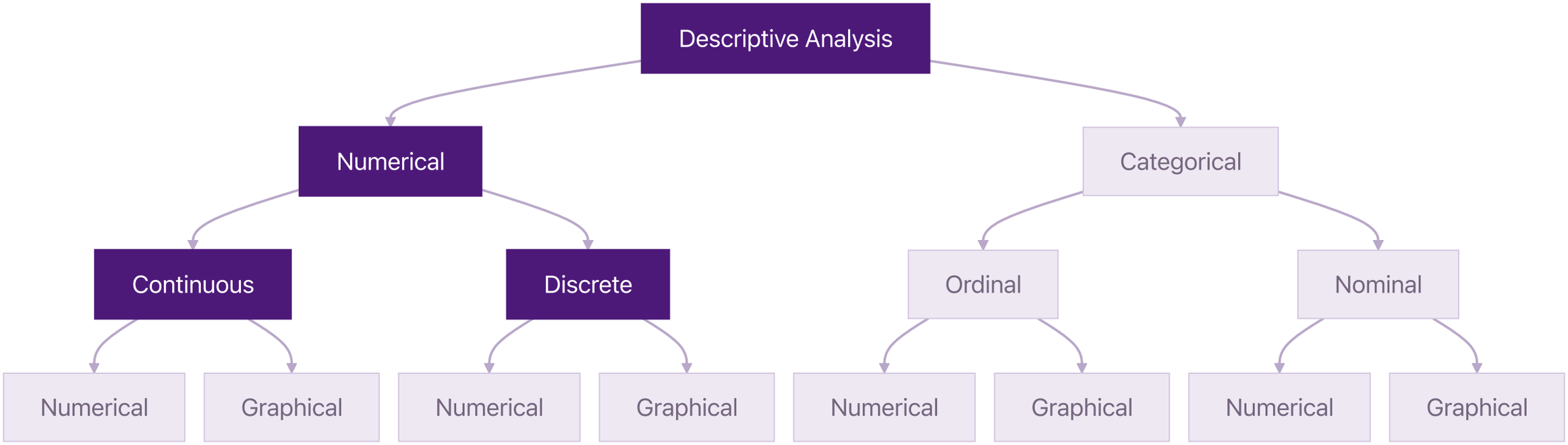
Continuous Data Summaries and Graphs

APHG 10203

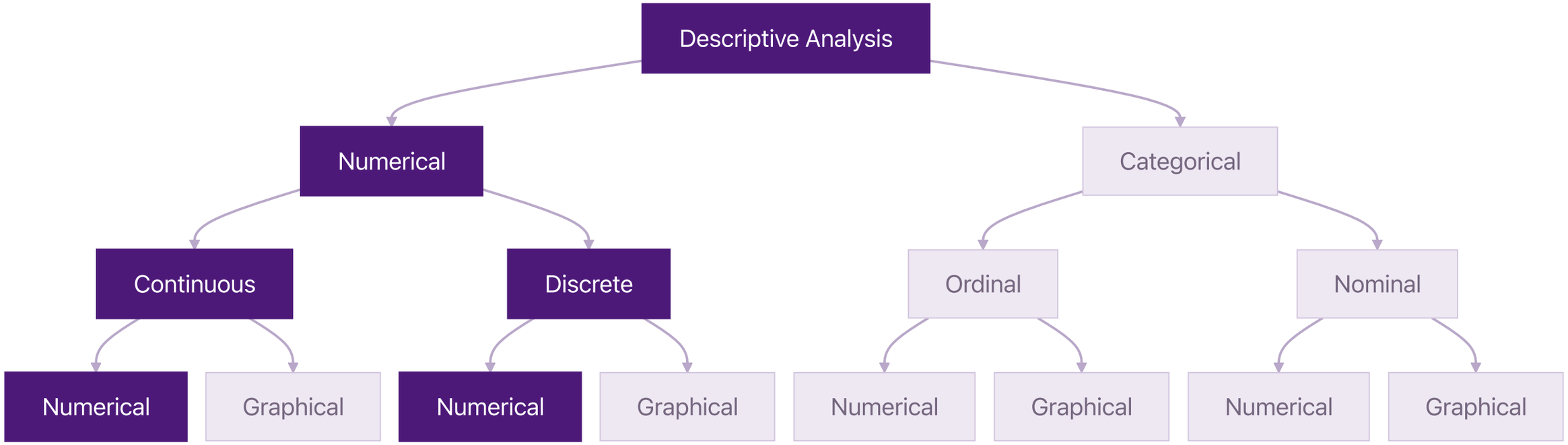
Descriptive Analysis



Numerical Data



Numerical Summaries



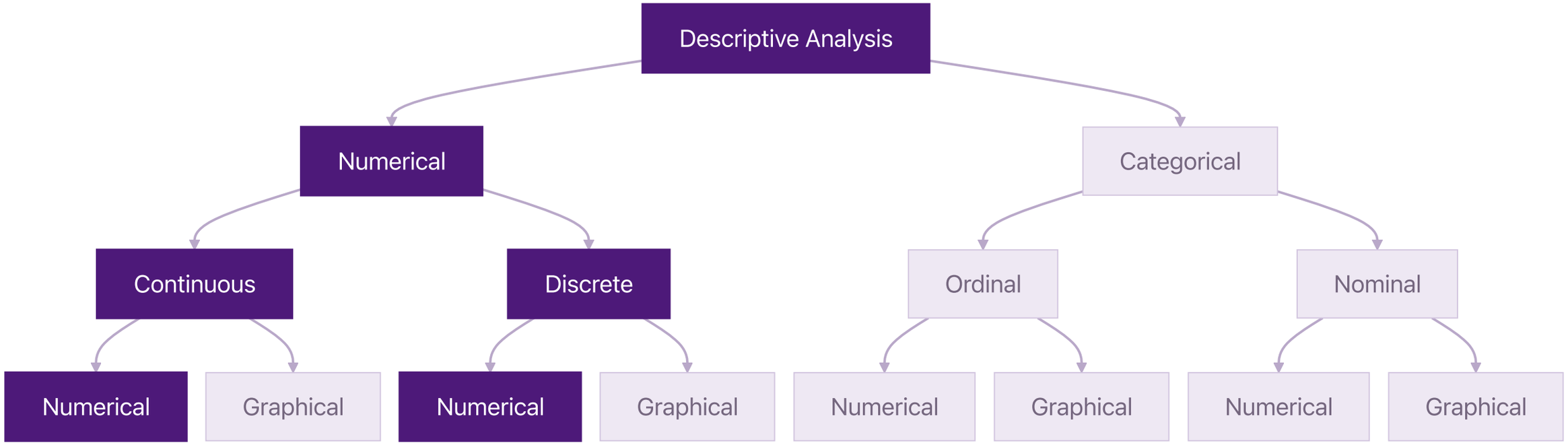
Review: 3 Important Characteristics of Numerical Variables

1. Center
2. Spread
3. Shape

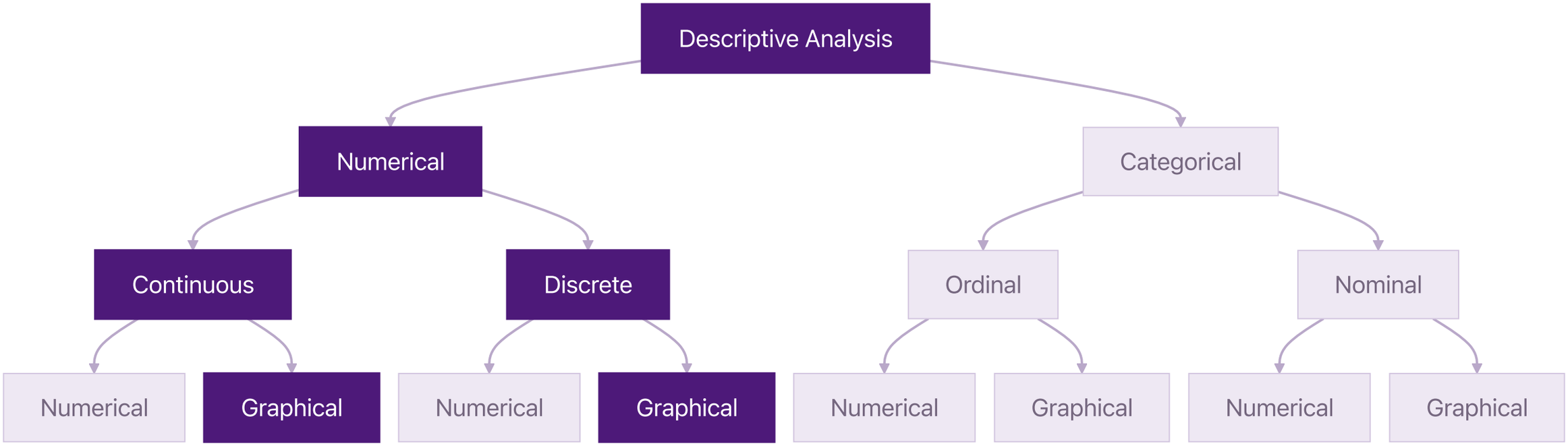
Review: 3 Measures of Central Tendency

1. Mean
2. Median
3. Mode

Numerical Summaries



Graphical Summaries



Review: Sleep Hours

Previous practice data: **12 fictional students' sleep hours**.

5, 5.5, 6, 6, 6.5, 6.5, 7, 7, 7.5, 8, 8, 9

- Why don't we stop at viewing/presenting raw data?
- What type of data is this?
- What three characteristics of this type of data should we typically summarize?
- What are the two **high-level methods** we can use to summarize this data?

Calculate the Mean in Excel

1. Download **practice_data.xlsx**.
2. Open **practice_data.xlsx**. Sleep hours are in **B2:B13**.
3. Type `Mean:` in cell **E2**.
4. Type `=AVERAGE(B2:B13)` in cell **F2**.

Calculate Measures of Central Tendency in Excel

Statistic	Excel formula	Result
Mean	`=AVERAGE(B2:B13)`	6.83 hours
Median	`=MEDIAN(B2:B13)`	
Mode		

Calculate Measures of Central Tendency in Excel

Statistic	Excel formula	Result
Mean	`=AVERAGE(B2:B13)`	6.83 hours
Median	`=MEDIAN(B2:B13)`	6.75 hours
Mode	`=MODE.MULT(B2:B13)`	

Calculate Measures of Central Tendency in Excel

Statistic	Excel formula	Result
Mean	`=AVERAGE(B2:B13)`	6.83 hours
Median	`=MEDIAN(B2:B13)`	6.75 hours
Mode	`=MODE.MULT(B2:B13)`	6, 6.5, 7, and 8 hours

Characteristics of Mean, Median, and Mode

Characteristic	Mean	Median	Mode
Simple	Yes	Yes	Yes
Only one	Yes	Yes	No
Susceptible to Extremes	Yes	No	No
Observed in Data	Not necessarily	Not necessarily	Yes, if exists

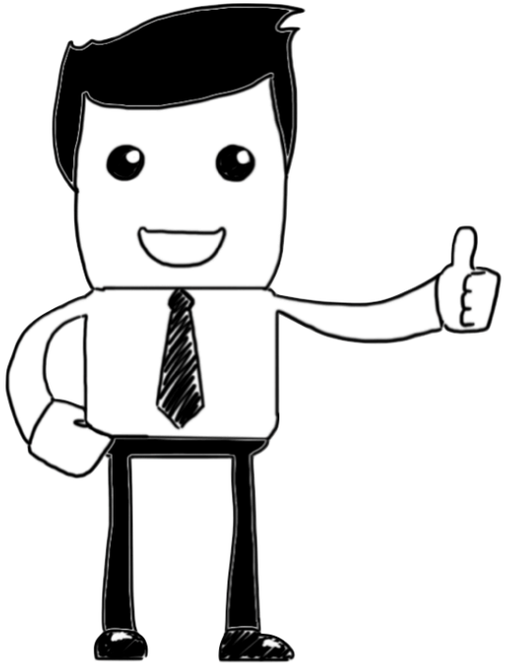
Measures of Dispersion

How Much Like the Average Is Everyone Else?

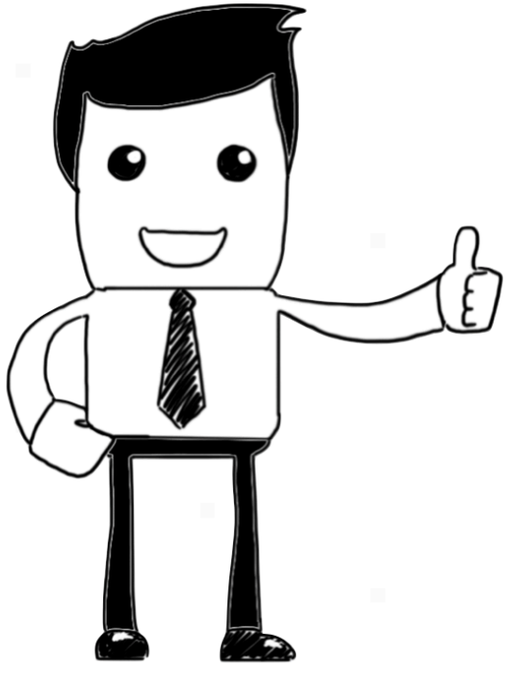
Mean Height = 68.4 Inches

How Much Like the Average Is Everyone Else?

68.4 Inches
5’ 8”



68.4 Inches
5’ 8”

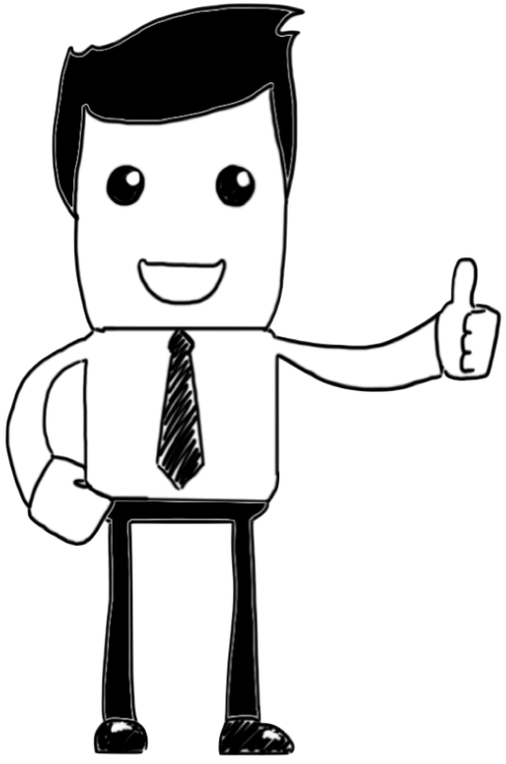


68.4 Inches
5’ 8”

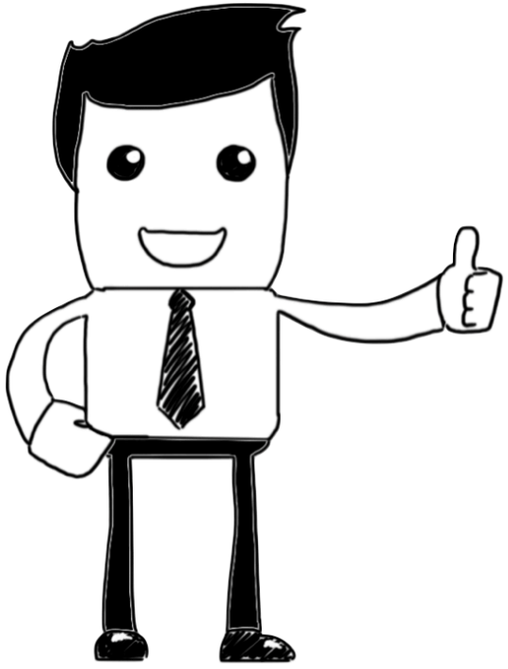


How Much Like the Average Is Everyone Else?

71 Inches
5’ 11”



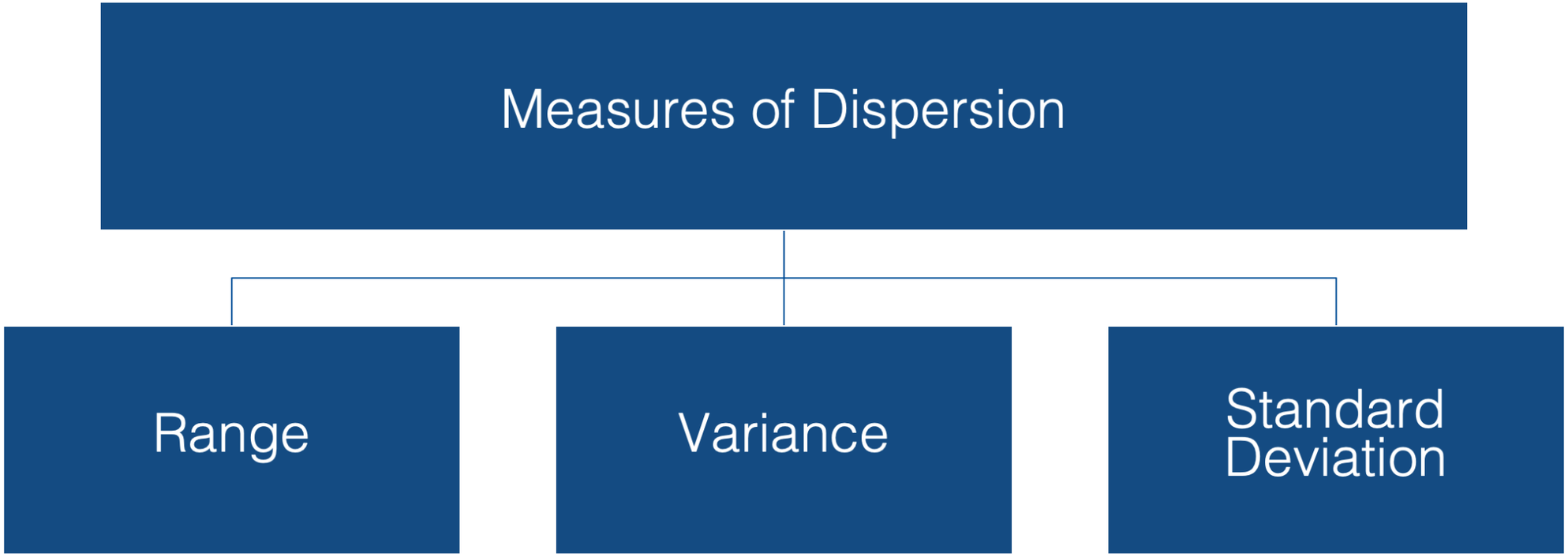
68.4 Inches
5’ 8”



64 Inches
5’ 4”

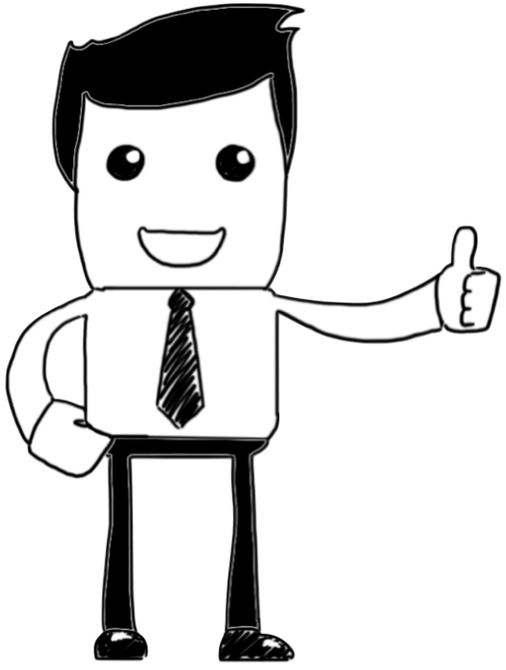


Measures of Dispersion

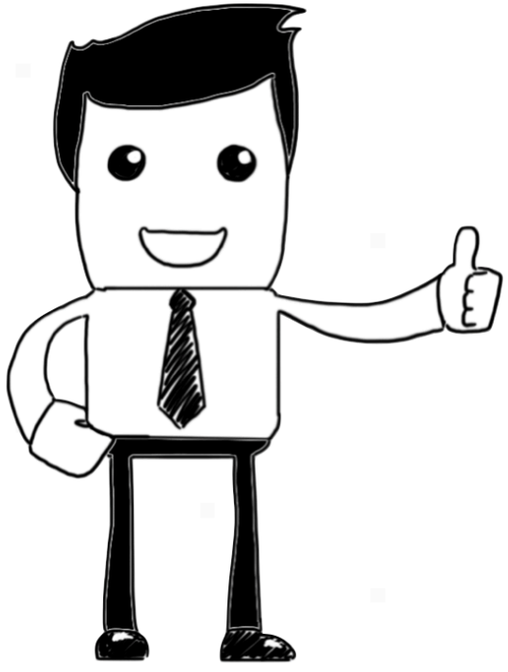


Range

68.4 Inches
5’ 8”



68.4 Inches
5’ 8”

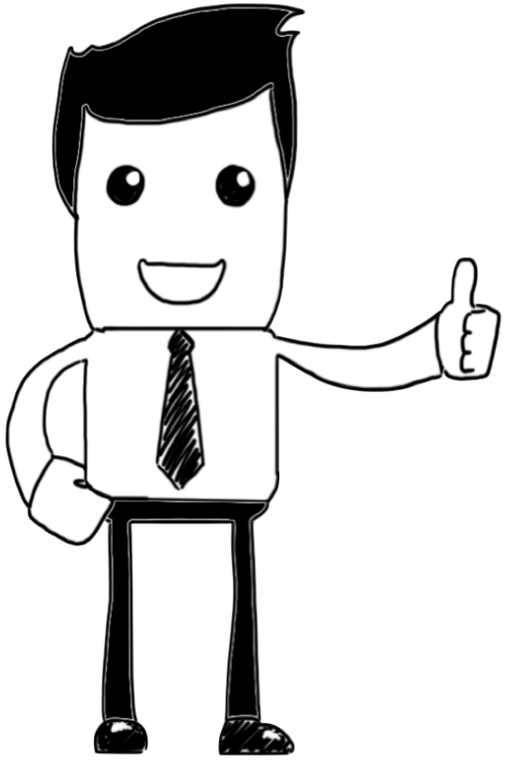


68.4 Inches
5’ 8”

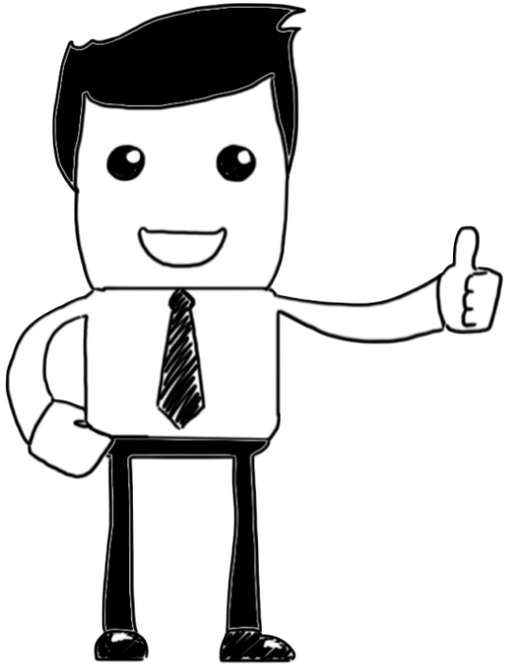


Range

71 Inches
5’ 11”



68.4 Inches
5’ 8”



64 Inches
5’ 4”



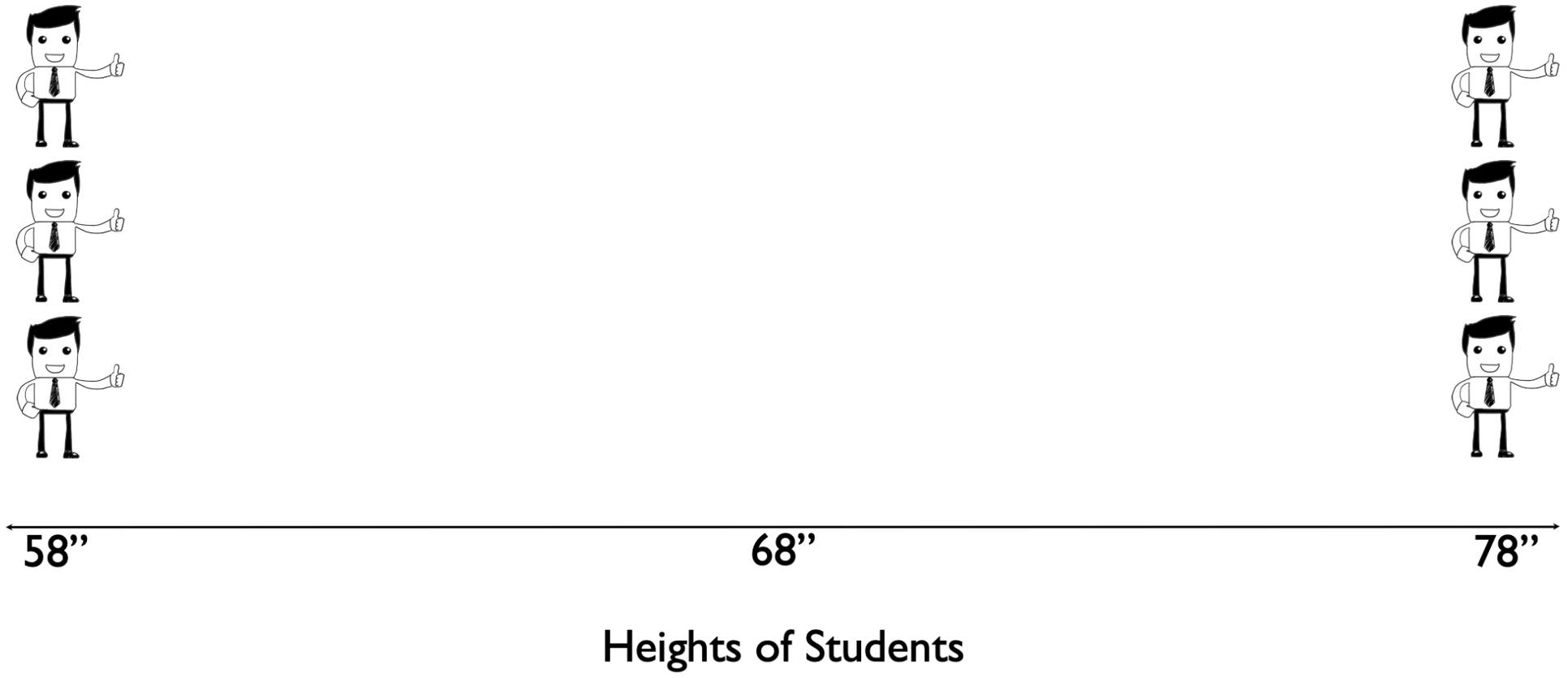
Calculate Measures of Dispersion in Excel

Statistic	Excel formula	Result
Min	`=MIN(C2:C21)`	
Max	`=MAX(C2:C21)`	
Range	`=MAX(C2:C21)-MIN(C2:C21)`	
Variance		
Standard Deviation		

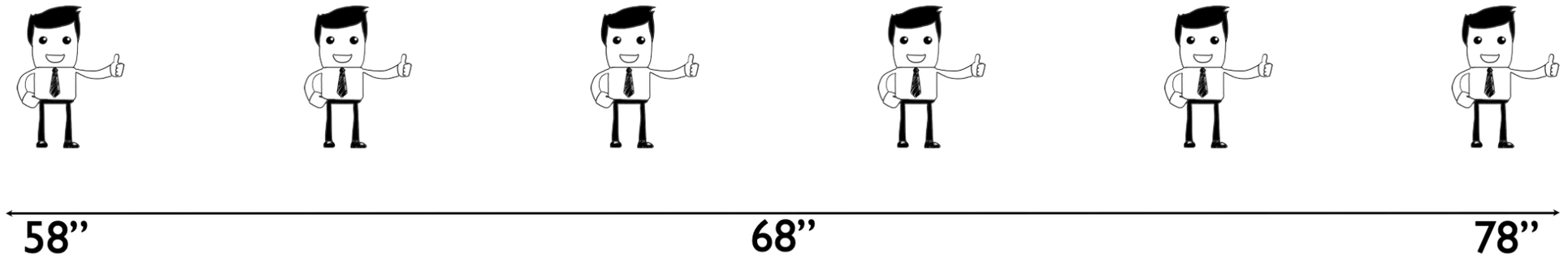
Calculate Measures of Dispersion in Excel

Statistic	Excel formula	Result
Min	`=MIN(C2:C21)`	64 inches
Max	`=MAX(C2:C21)`	75 inches
Range	`=MAX(C2:C21)-MIN(C2:C21)`	11 inches
Variance		
Standard Deviation		

How Much Like the Average Is Everyone Else?

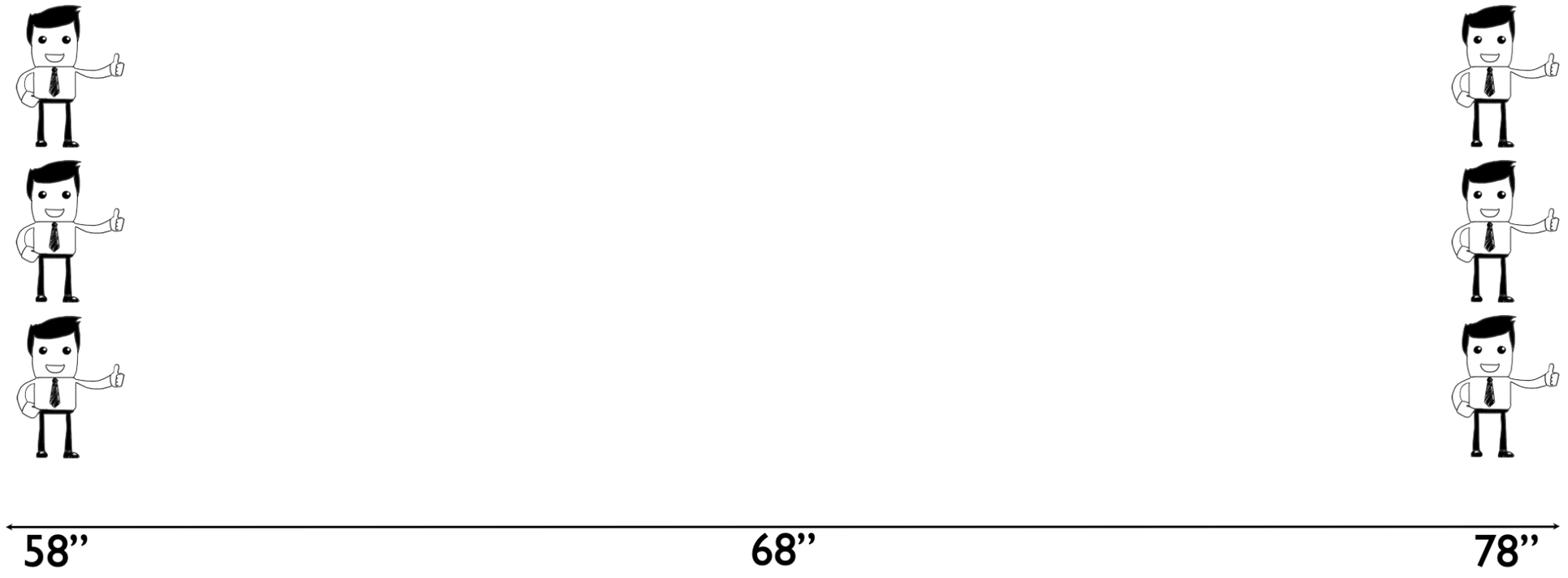


How Much Like the Average Is Everyone Else?

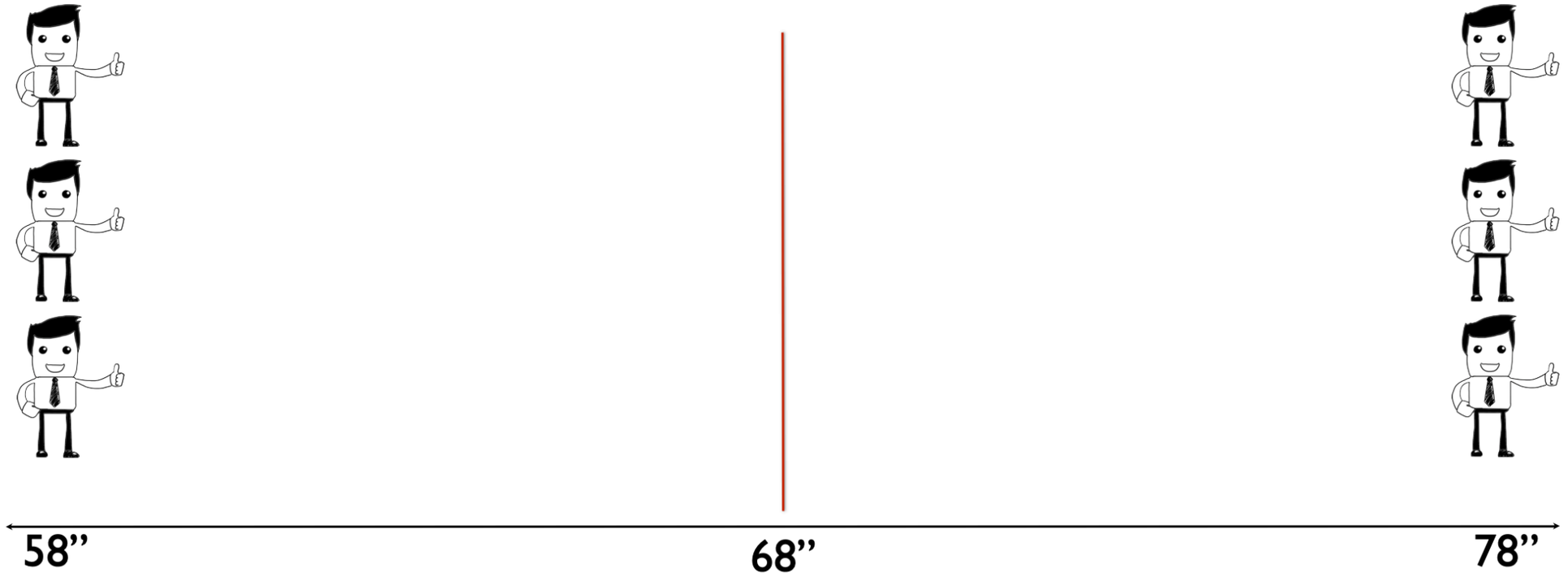


Heights of Students

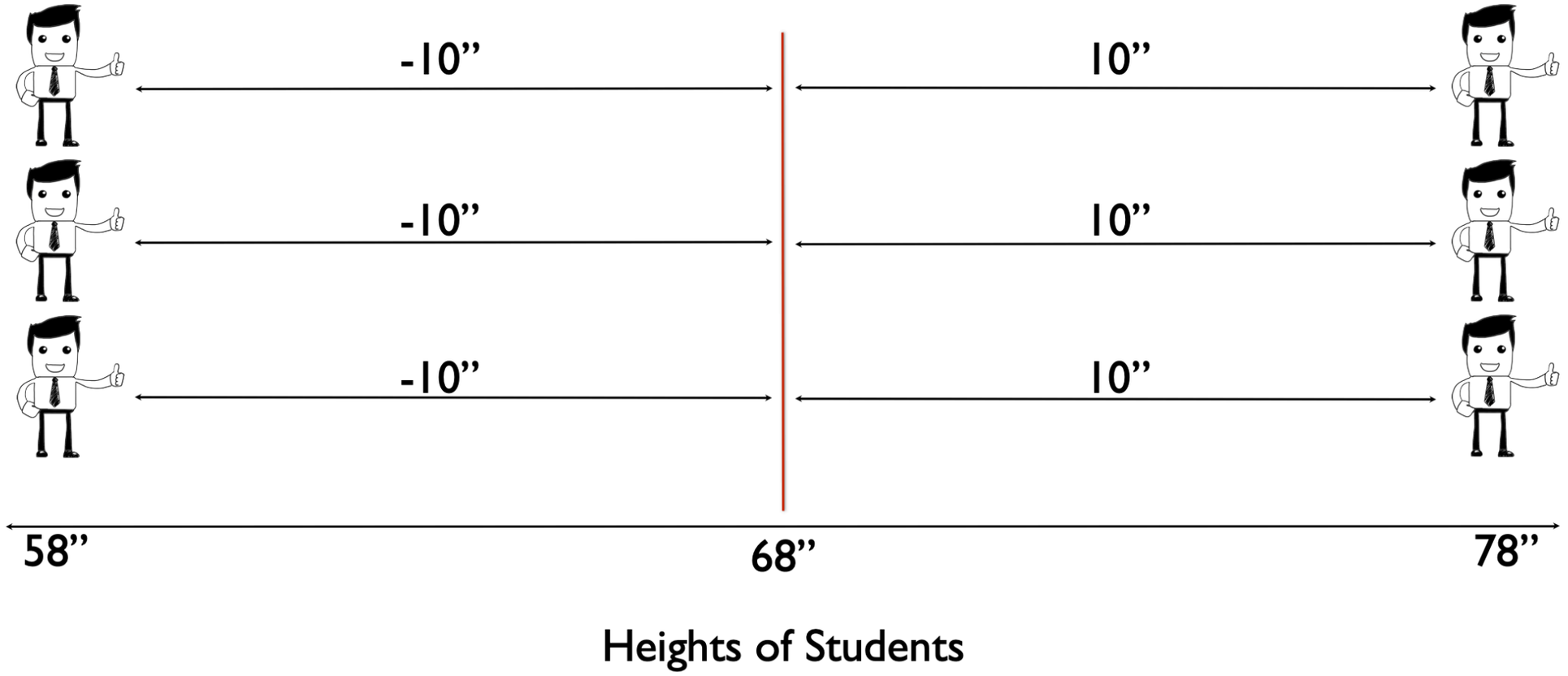
Variance



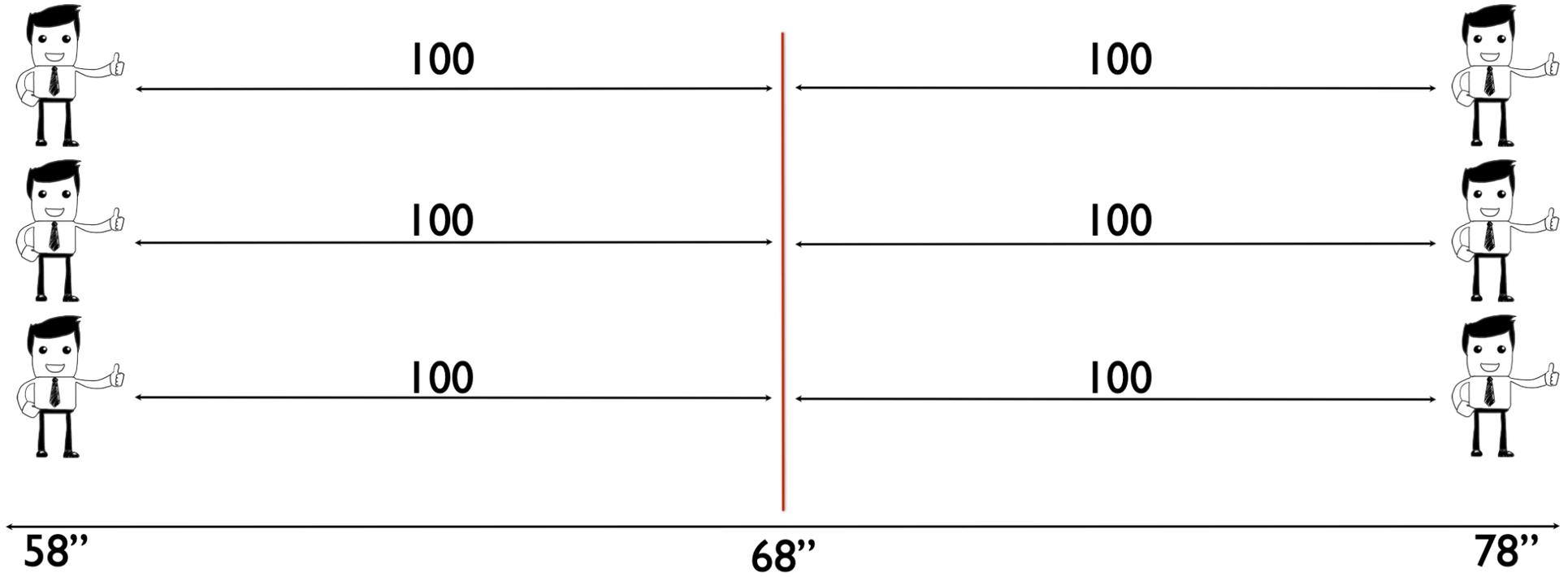
Variance



Distances from the Mean

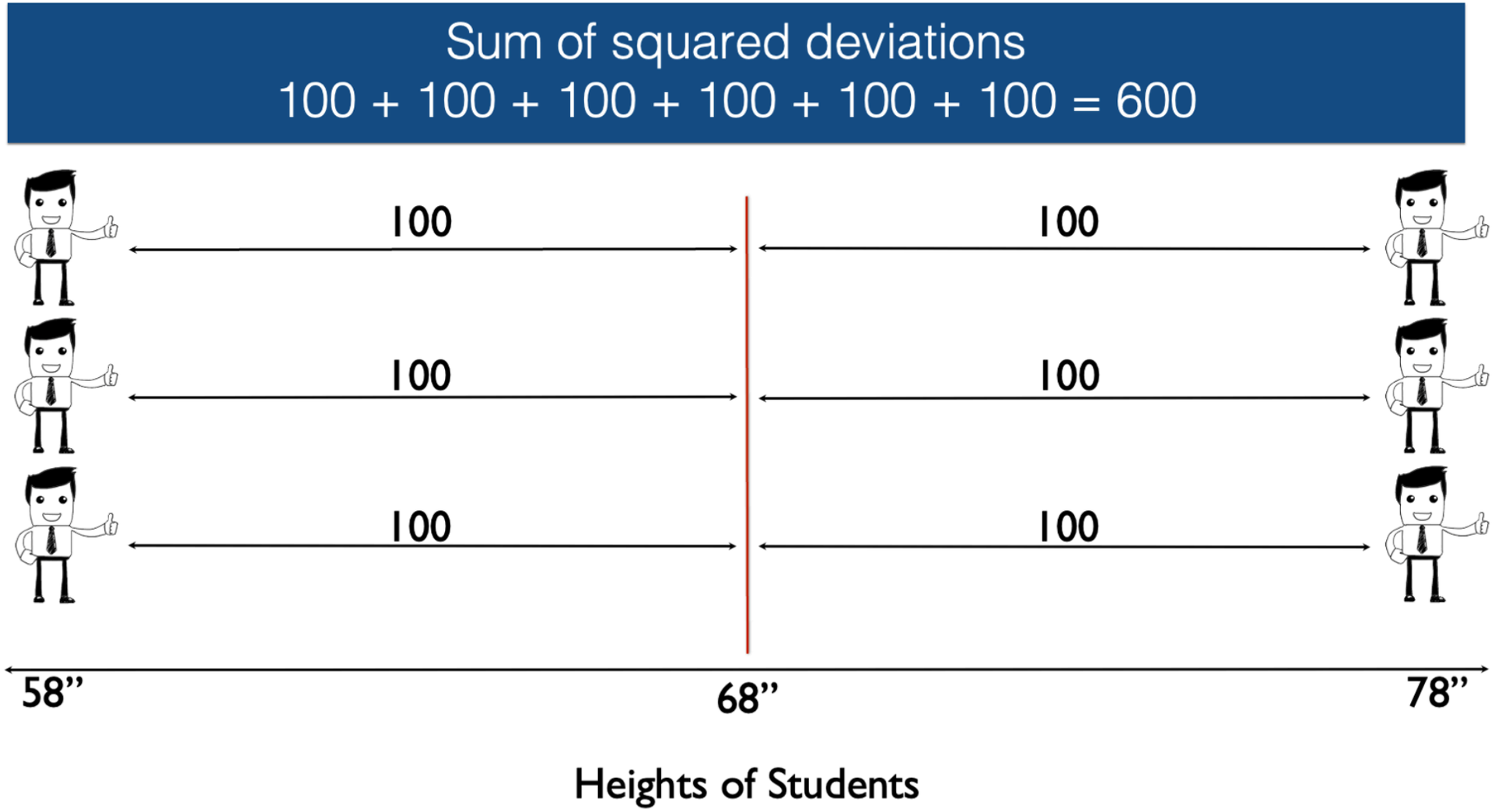


Square the Differences



Heights of Students

From Squared Distances to Variance



Variance

$s^2 = \frac{600}{6-1} = 120$ **square inches**.

Calculate Measures of Dispersion in Excel

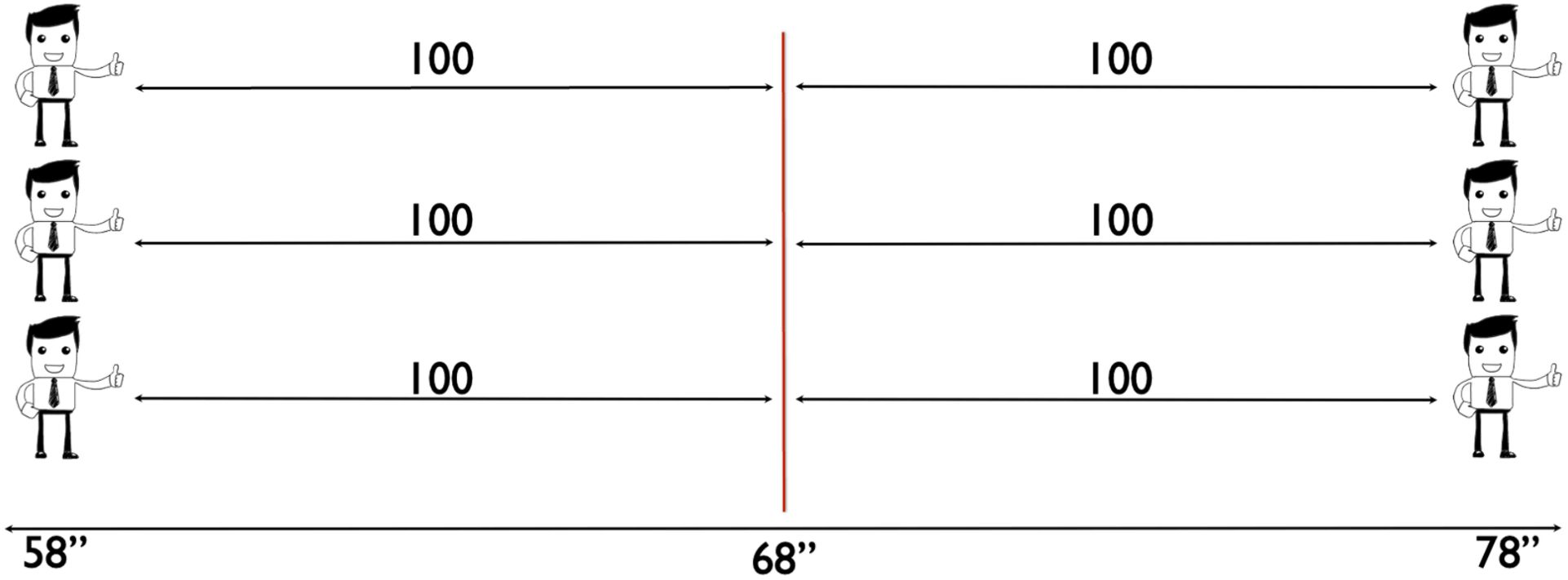
Statistic	Excel formula	Result
Min	`=MIN(C2:C21)`	64 inches
Max	`=MAX(C2:C21)`	75 inches
Range	`=MAX(C2:C21)-MIN(C2:C21)`	11 inches
Variance	`=VAR.S(C2:C21)`	
Standard Deviation		

Calculate Measures of Dispersion in Excel

Statistic	Excel formula	Result
Min	`=MIN(C2:C21)`	64 inches
Max	`=MAX(C2:C21)`	75 inches
Range	`=MAX(C2:C21)-MIN(C2:C21)`	11 inches
Variance	`=VAR.S(C2:C21)`	10.04 square inches
Standard Deviation		

Standard Deviation Returns to the Original Units

Standard Deviation $(s) = \sqrt{s^2} = \sqrt{120} \approx 10.95$



Heights of Students

Calculate Measures of Dispersion in Excel

Statistic	Excel formula	Result
Min	`=MIN(C2:C21)`	64 inches
Max	`=MAX(C2:C21)`	75 inches
Range	`=MAX(C2:C21)-MIN(C2:C21)`	11 inches
Variance	`=VAR.S(C2:C21)`	10.04 square inches
Standard Deviation	`=STDEV.S(C2:C21)`	

Calculate Measures of Dispersion in Excel

Statistic	Excel formula	Result
Min	`=MIN(C2:C21)`	64 inches
Max	`=MAX(C2:C21)`	75 inches
Range	`=MAX(C2:C21)-MIN(C2:C21)`	11 inches
Variance	`=VAR.S(C2:C21)`	10.04 square inches
Standard Deviation	`=STDEV.S(C2:C21)`	3.17 inches

Exercise: Comparing Distributions

1. Open **practice_data.xlsx**.
2. Select the **comparing_distributions** sheet.
3. Complete the **Comparisons** table.

Z-Scores: Distance in SD Units

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$$

Z-score	Interpretation
Positive	Above the mean
0	At the mean
Negative	Below the mean

Check: Calculate and Interpret a Z-Score

A fictional score distribution has **mean 30** and **SD 5**.

What is the z-score for a value of 20?

$$z = \frac{20 - 30}{5} = -2$$

The value is **2 standard deviations below the mean**.

Z-Scores: Why?

Health assessment	Maria's score	Reference mean	SD	Z-score
Physical functioning	85	75	5	+2
Emotional well-being	90	80	10	+1

Quartiles and the Middle Half

Position	Meaning	Sleep hours
Q1	25th percentile	6.00 hours
Q2	50th percentile (Median)	6.750 hours
Q3	75th percentile	7.625 hours
IQR	Q3 – Q1	1.625 hours

Calculate Quartiles in Excel

Position	Excel formula	Result
Q1	`=QUARTILE.INC(C2:C21,1)`	
Q2	`=QUARTILE.INC(C2:C21,2)`	
Q3	`=QUARTILE.INC(C2:C21,3)`	
IQR	`=G13-G11` or	
IQR	`=QUARTILE.INC(C2:C21,3) - QUARTILE.INC(C2:C21,1)`	

Calculate Quartiles in Excel

Position	Excel formula	Result
Q1	`=QUARTILE.INC(C2:C21,1)`	65.75 inches
Q2	`=QUARTILE.INC(C2:C21,2)`	68.5 inches
Q3	`=QUARTILE.INC(C2:C21,3)`	71 inches
IQR	`=G13-G11` or	5.25 inches
IQR	`=QUARTILE.INC(C2:C21,3) - QUARTILE.INC(C2:C21,1)`	5.25 inches

Excel: Numerical Summaries

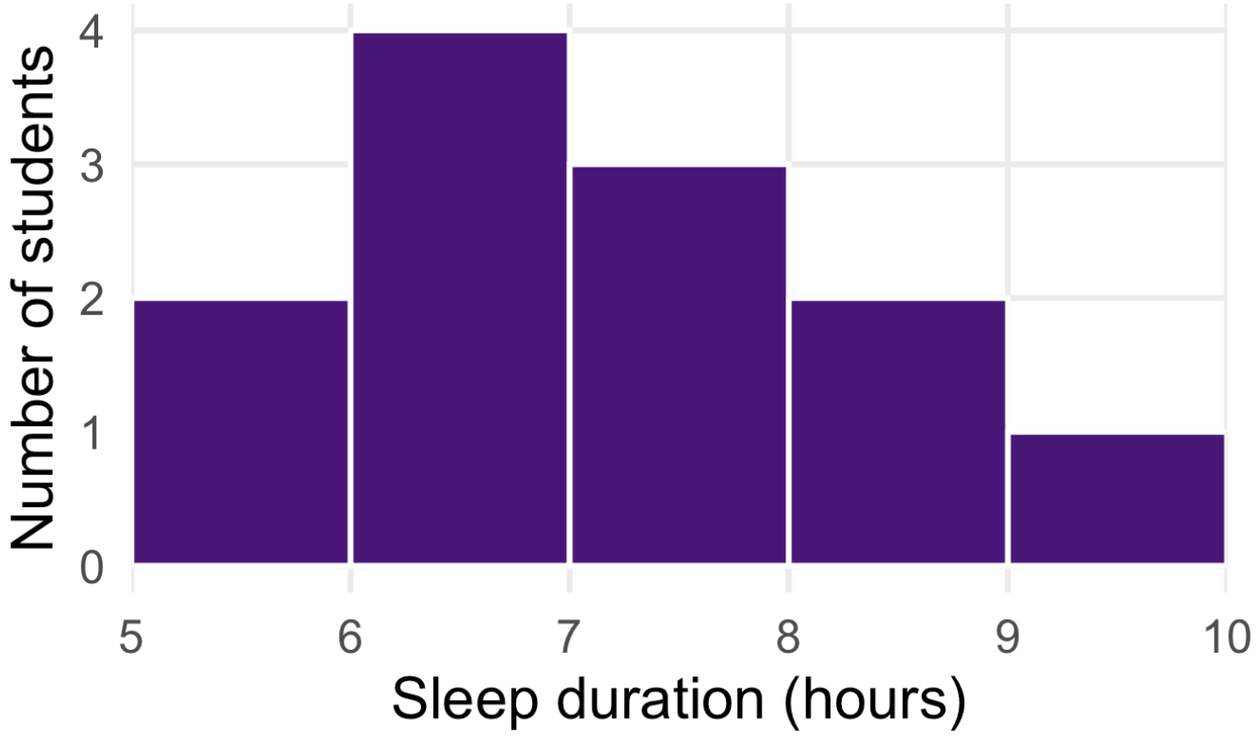
Summary	Formula for sleep_hours	Expected result
n	`=COUNT(B2:B13)`	12
Mean	`=AVERAGE(B2:B13)`	6.83
Median	`=MEDIAN(B2:B13)`	6.75
Mode	`=MODE.MULT(B2:B13)`	6, 6.5, 7, and 8
Minimum	`=MIN(B2:B13)`	5
Maximum	`=MAX(B2:B13)`	9
Range	`=MAX(B2:B13)-MIN(B2:B13)`	4
Sample variance	`=VAR.S(B2:B13)`	1.33
Sample SD	`=STDEV.S(B2:B13)`	1.15
Q1	`=QUARTILE.INC(B2:B13,1)`	6.00
Q2	`=QUARTILE.INC(B2:B13,2)`	6.75
Q3	`=QUARTILE.INC(B2:B13,3)`	7.62
IQR	`=QUARTILE.INC(B2:B13,3)-QUARTILE.INC(B2:B13,1)`	1.62

Graphical Summaries

Choosing a Chart/Graph for One Variable

Variable Type	Usually Appropriate
Categorical	Bar chart Pie chart
Continuous Numerical	Histogram Boxplot Density plot
Discrete Numerical	Bar chart Histogram Dot plot

Histograms

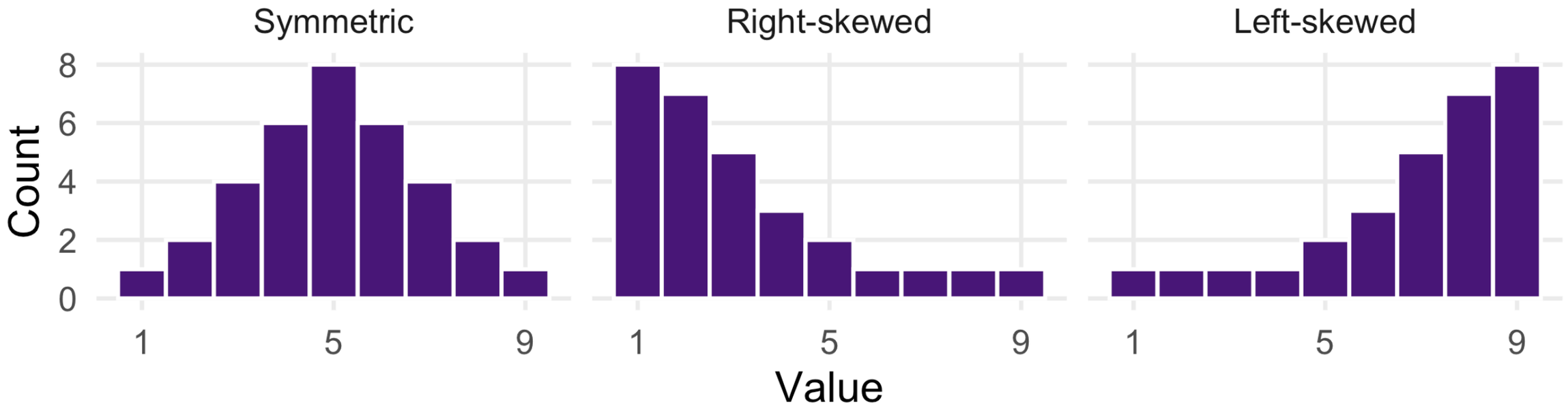


A histogram groups numerical values into intervals called **bins**.

- The horizontal axis shows the values and their units.
- The height of each bar shows how many observations fall in that bin.
- The bars touch because the bins cover adjacent intervals.

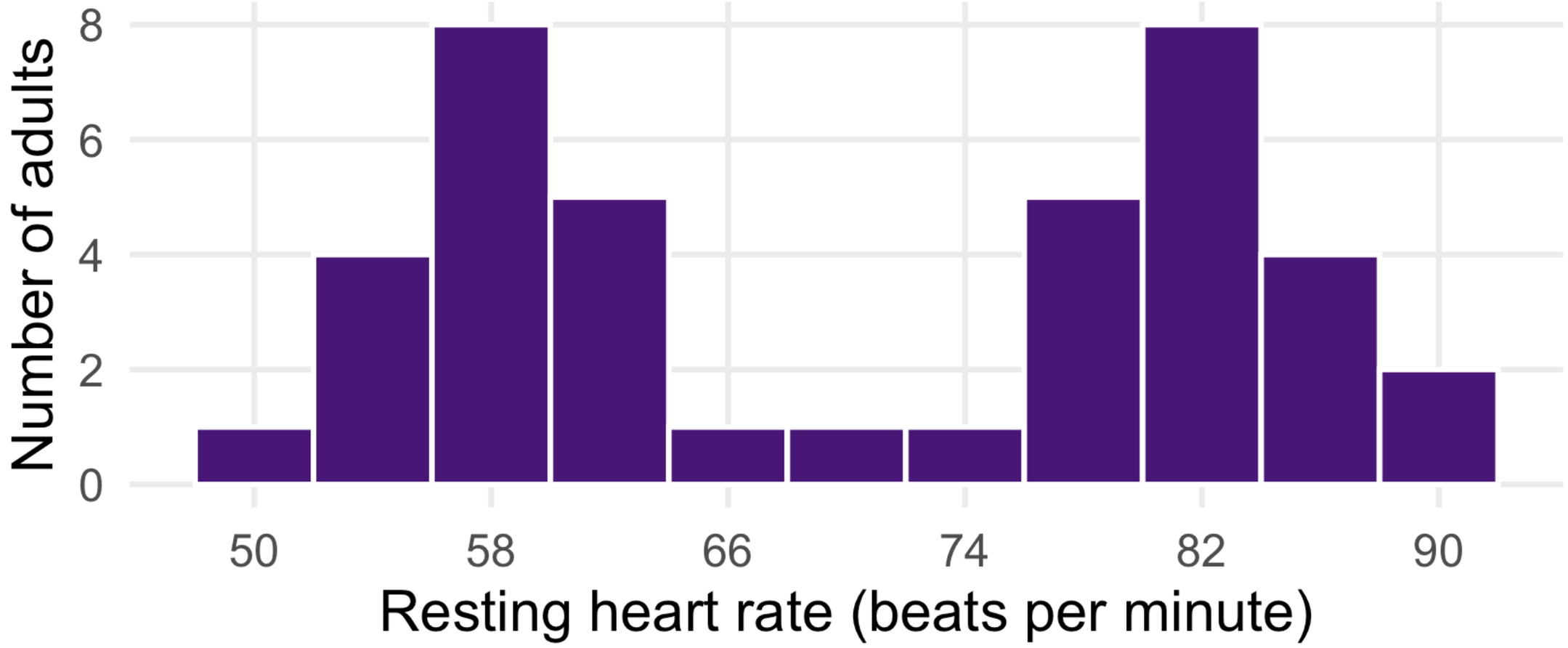
Use the pattern of bars to examine the distribution's shape and spread.

Reading Distribution Shape in a Histogram



Name skew by the direction of the longer tail.

A Bimodal Distribution Has Two Peaks



Bimodal means that the distribution has two distinct peaks.

Check: Which Direction Is the Skew?

Most clinic visits last **15-25 minutes**, but a few last **60-90 minutes**.

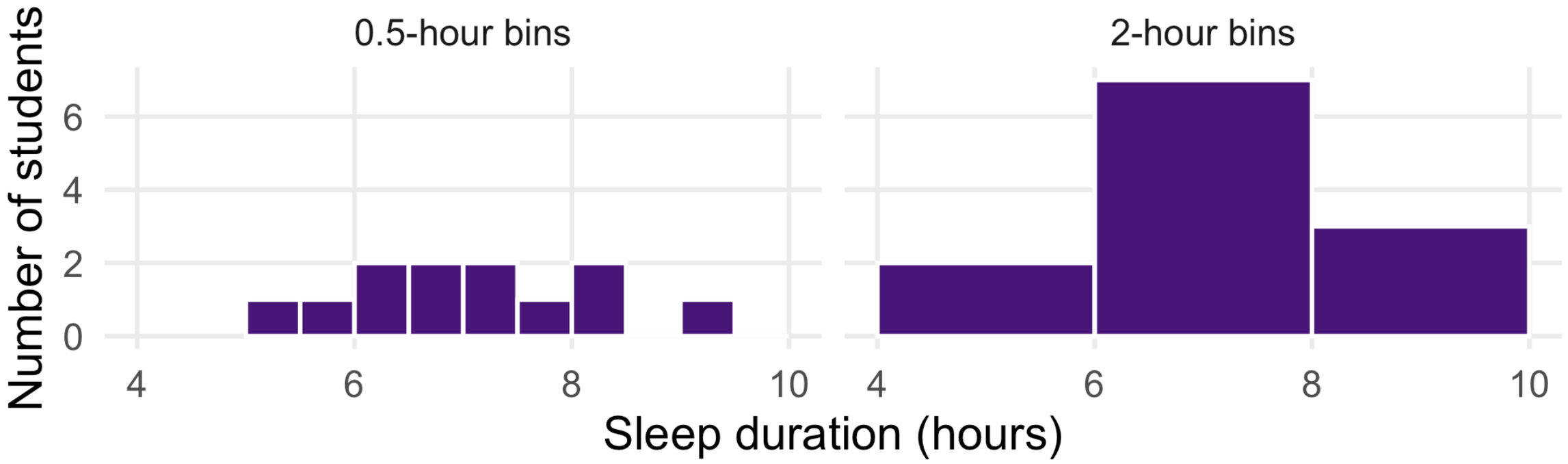
A. Left-skewed

B. Right-skewed

C. The long visits make the distribution symmetric

B. Right-skewed. The long visits extend the tail toward larger values.

Histogram: Bin Width Changes the Picture



Did the students' sleep change, or only how we grouped it?

Only the grouping changed. Inspect more than one sensible bin width.

Create a Histogram in Excel

1. Select **B1:B13**, including the sleep_hours header.
2. Choose **Insert → Statistic Chart → Histogram**.
3. Add a clear title and axis titles: **Sleep duration (hours)** and **Number of students**.
4. Inspect the bins. Try a **1-hour bin width** and compare with the automatic grouping.

Do the plotted counts add up to 12?

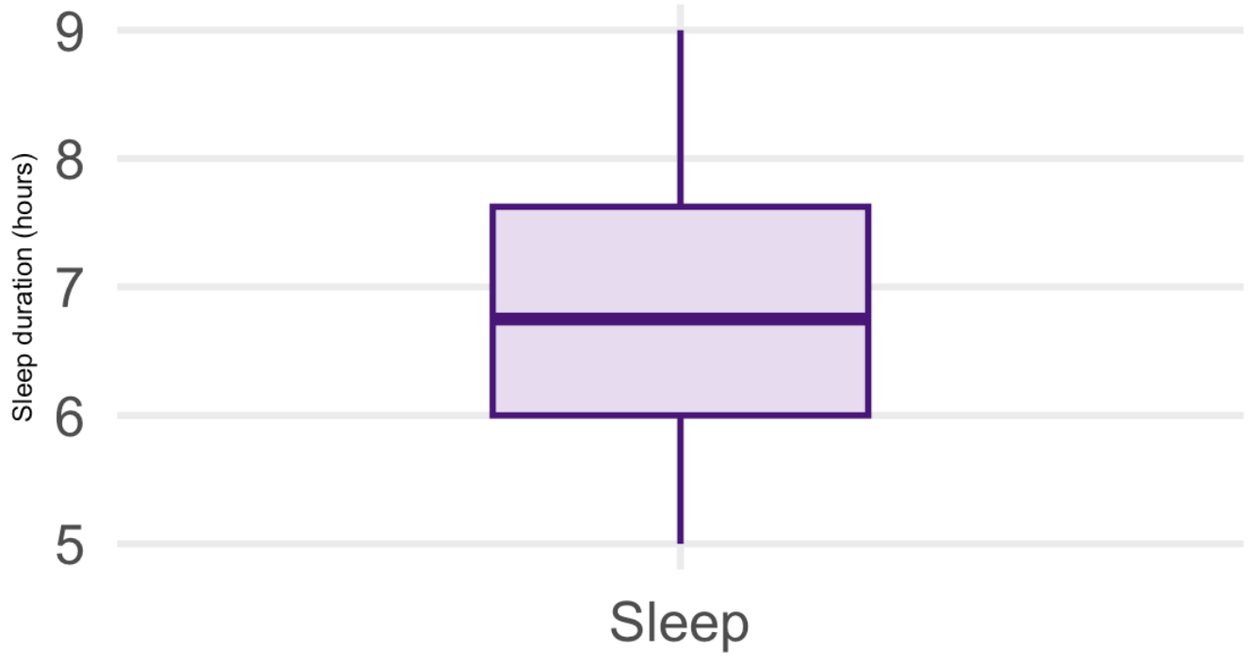
Describe What You Can See

A useful histogram description addresses:

- **Shape:** where values cluster and which direction any tail extends.
- **Spread:** the observed values and units.
- **Possible outliers:** observations that stand apart and warrant checking.

Boxplots

Reading a Boxplot



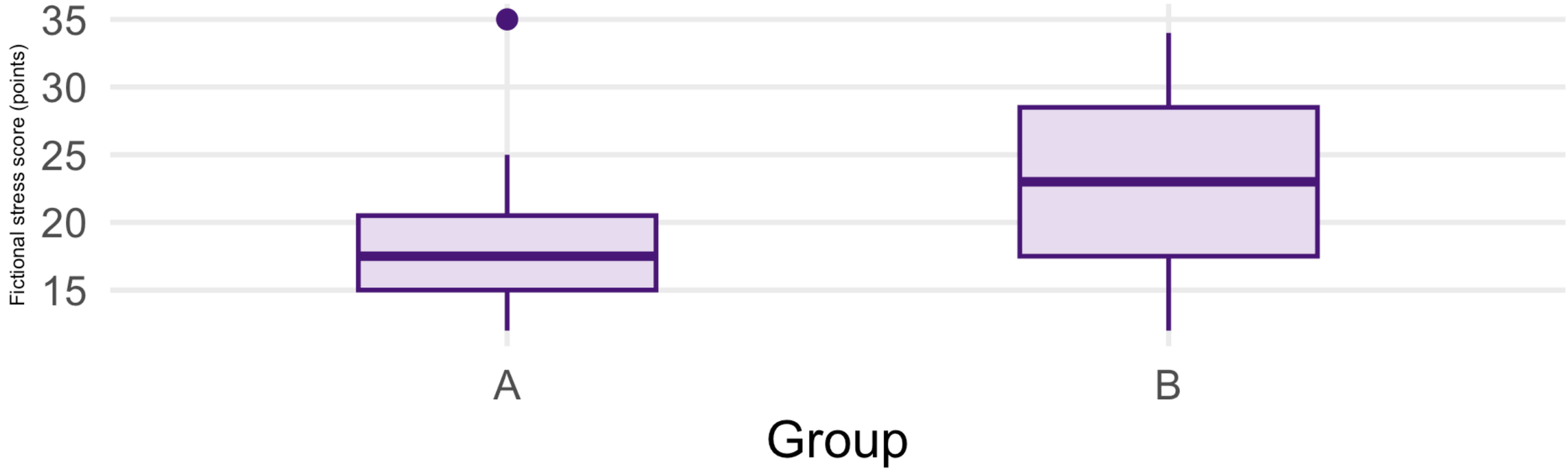
- Line inside the box: **median**.
- Lower and upper box edges: **Q1 and Q3**. Box height: **IQR**.
- Whiskers and separate points show values beyond the middle half.

Whiskers and Potential Outliers

For the boxplots in this lesson:

- Fences are **Q1 – 1.5 × IQR** and **Q3 + 1.5 × IQR**.
- Whiskers reach the most extreme observed values **within the fences**.
- Values beyond the fences appear as separate points.

Check: Compare These Boxplots



Which group has the higher median? Larger IQR? A flagged outlier?

B: higher median and larger IQR. **A:** a flagged point at 35.

Creating a Boxplot in Excel

1. Select **B1:B13**, including the sleep_hours header.
2. Choose **Insert → Statistic Chart → Box & Whisker**.
3. Change the chart title to **Sleep Duration** and add the vertical axis title **Hours**.
4. Right-click the box and choose **Format Data Series**. Select **Show inner points** to display each student's value.

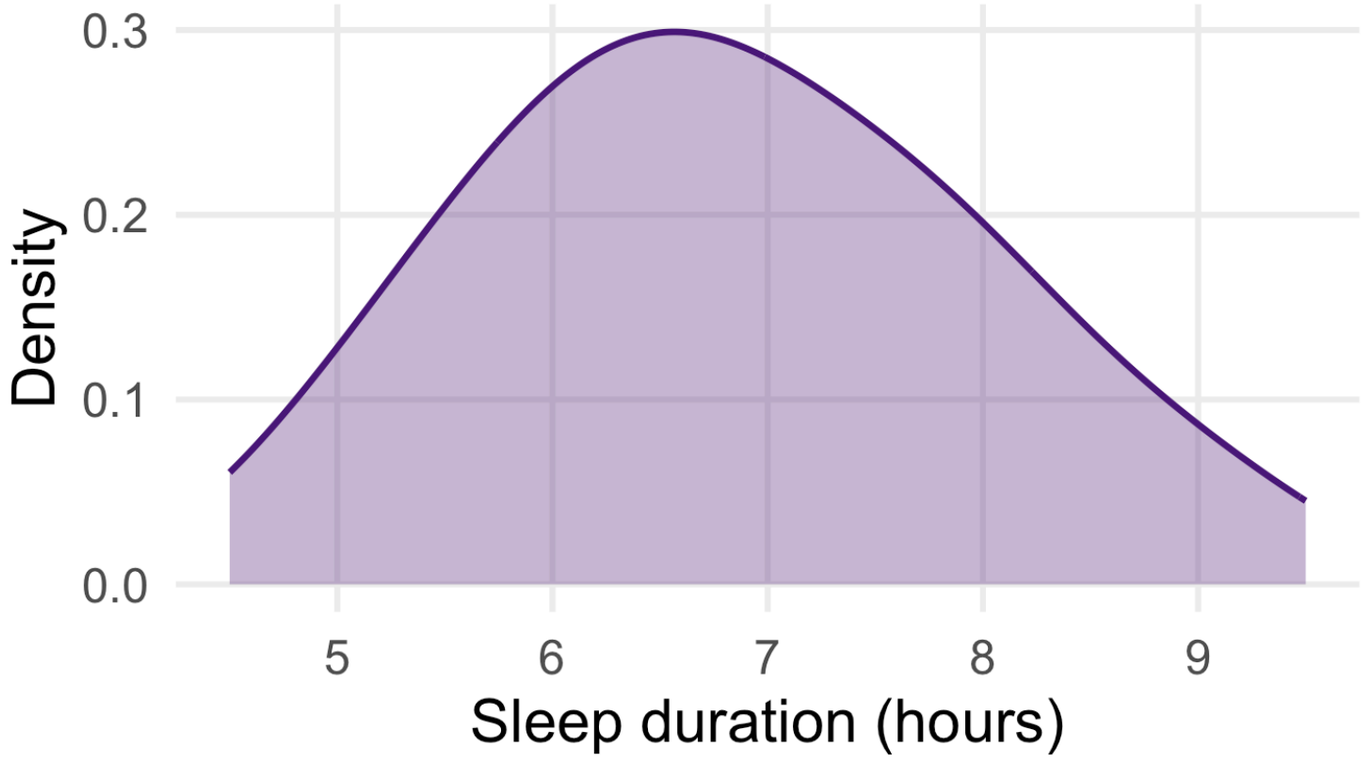
Find the median and IQR. Does Excel flag any potential outliers?

Median = **6.75 hours**; IQR = **1.625 hours**. Excel flags **no potential outliers**.

Bar Chart

1. Open the **height_and_weight** sheet and select **A1:B21**, including the headers.
2. Choose **Insert → PivotTable**. Place the PivotTable in a new worksheet.
3. Drag **sex** to **Rows** and **id** to **Values**. Confirm that Values displays **Count of id**.
4. Select the PivotTable and choose **Insert → Bar Chart → 2-D Clustered Bar**.
5. Title the chart **Students by Sex**. Label the horizontal axis **Number of students**.

Density Plot

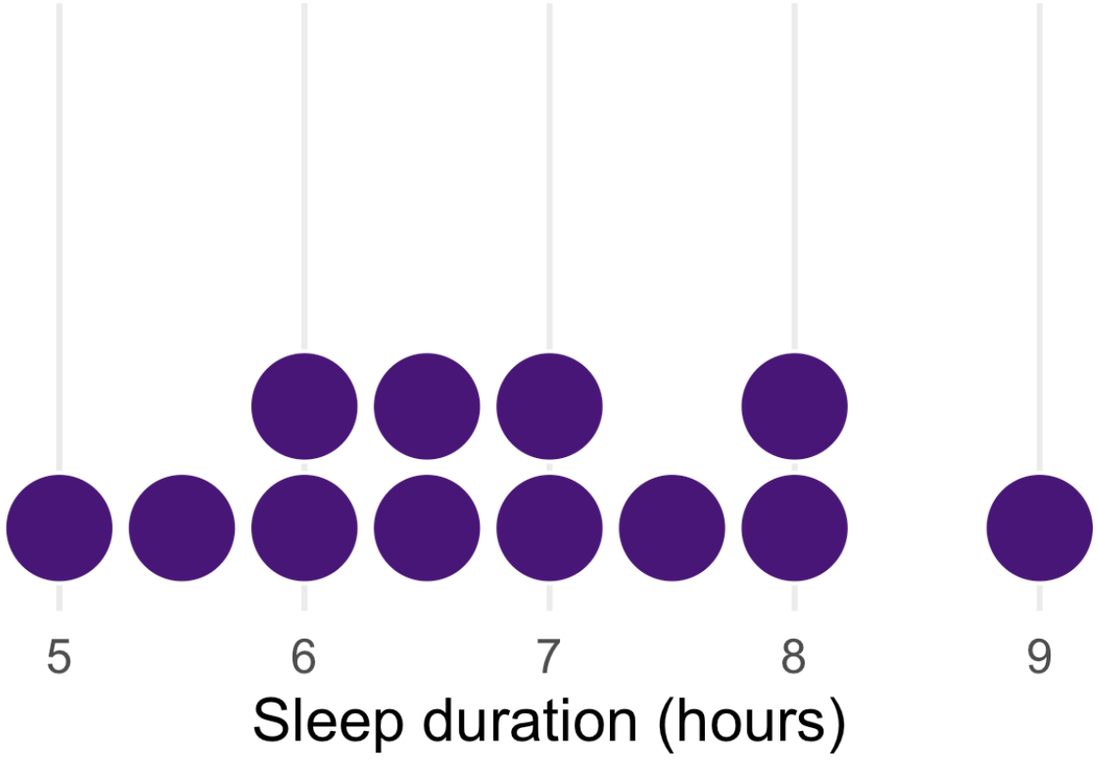


A density plot shows a **smoothed estimate** of a distribution's shape.

- Higher parts of the curve indicate where observations are more concentrated.
- The total area under the curve equals 1.
- The amount of smoothing affects the curve.

With only 12 observations, use this plot as a rough summary.

Dot Plot



A dot plot displays the **individual observations**.

- Each dot represents one student.
- Dots stack when students report the same value.
- Gaps and repeated values remain visible.

Dot plots work especially well for small datasets.

Group Exercise: Describe Stress Scores

Use **stress_score in C2:C13** from the same practice file.

1. Calculate **n, mean, sample SD, minimum, and maximum**.
2. Create a histogram with a title and labeled axes.
3. Write three observations about **shape, spread, and possible outliers**.
4. Be ready to explain one number and one feature of your graph.

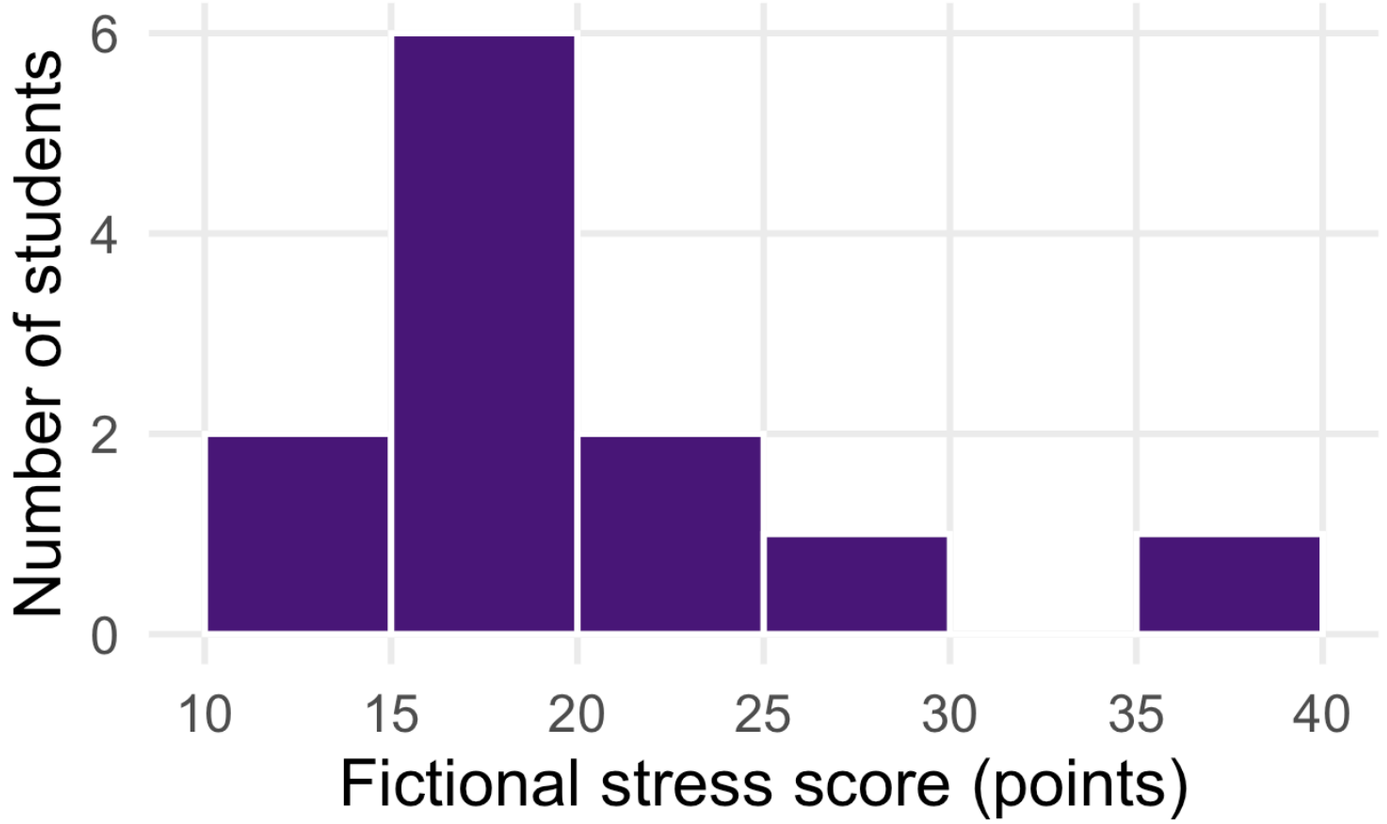
Work for 8 minutes. We will use 4 minutes to compare answers.

Practice Data and Graph

Stress scores, sorted:

12, 14, 15, 15, 16, 17, 18, 19, 20, 22, 25, 35

Describe the distribution in your own words.



Group Exercise: Compare Your Results

Summary	stress_score
n	12
Mean	19.00 points
Sample SD	6.19 points
Minimum; maximum	12; 35 points

- Most scores lie between **12 and 25**, with a tail toward higher scores.
- Observed scores span **23 points**.
- **35** stands apart and warrants checking; it is not automatically an error.

Connection to Prep 3

For your **assigned outcome**, report:

- **n, mean, SD, minimum, and maximum**.
- A labeled histogram and observations about its shape, spread, and possible outliers.

Use your assigned project dataset, not today's practice file.

Scatterplots come in the next lesson. Prep 3 also requires variable classifications and individual contribution statements.

Sources

- Brad Cannell. *Measures of Central Tendency* and *Measures of Dispersion*, Introduction to Univariate Descriptive Analysis in R. Adapted explanations and height examples.
- Yan Zhang. *Statistics Overview and Descriptive Statistics*, APHS 20203. Selected concepts adapted from slides 30-33 and 36-49 of the supplied PowerPoint.
- APHS 20203: *Statistical Thinking and Describing Data*, the approved course outline, and *Group Presentation - Prep 3*.
- Microsoft Support: Create a histogram and STDEV.S.

All numerical examples are **fictional teaching examples**.